

Problem Set 1 [Applied Stats/Quant Methods 1]

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Question 1: Education

1.

```
y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98, 80, 97, 95, 111, 114, 89)
```

number of observations

```
n = length(y)
```

descriptive statistics

```
mean_y <- sum(y)/n
sd_y <- sqrt(sum((y - mean_y) ^ 2) / (n - 1))
se_y <- sd_y/sqrt(n)
t_score <- qt(0.95, df = n - 1)

ci_90_lower = mean_y - t_score * se_y
ci_90_upper = mean_y + t_score * se_y
```

hypothesis test

variables:

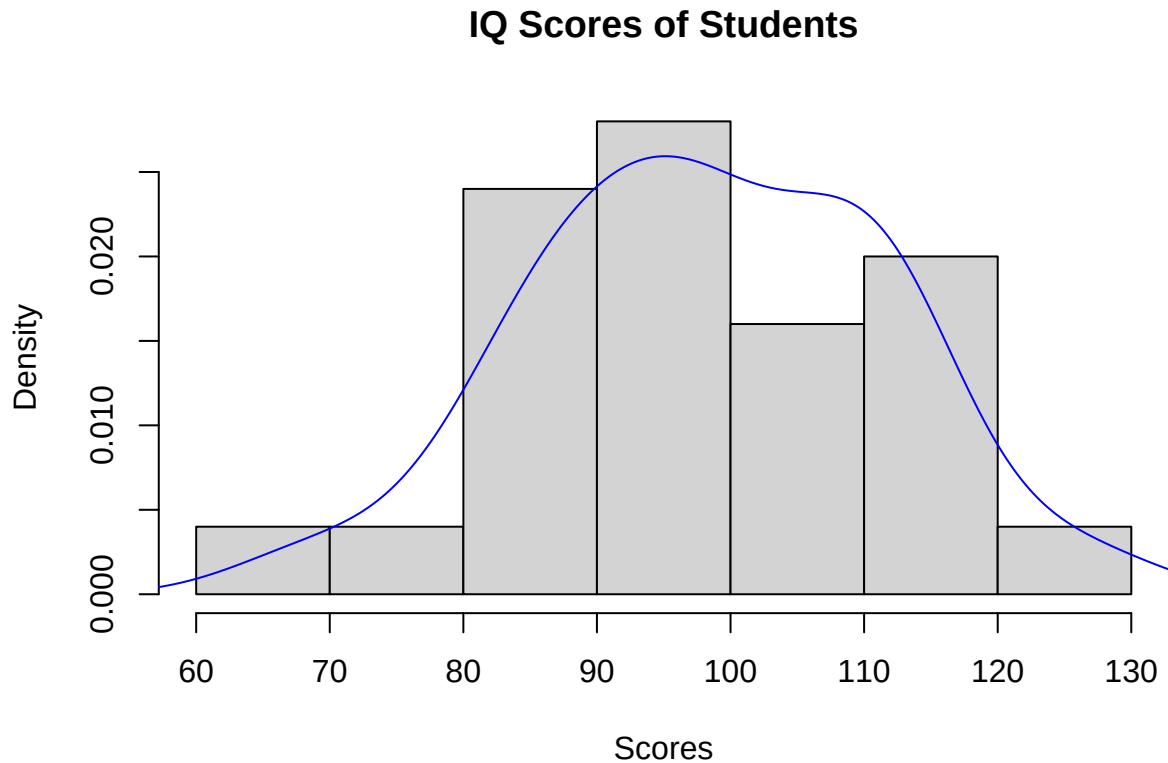
explanatory: students (numeric)

response: IQ scores (numeric)

visualizing the distribution

```
hist(y,
     breaks = 8,
     probability = TRUE,
     main = "IQ Scores of Students",
     xlab = "Scores")

lines(density(y), col = "blue")
```



2.

significance tests

question: Is the average IQ score in the counselor's school from the sample higher than the average IQ score (100) among all the schools in the country?

hypotheses:

H0: average IQ score of the students is 100 ($\mu = 100$)

HA: average IQ score of the students is greater than 100 ($\mu > 100$)

```
t.test(y, mu = 100, alternative = "greater", conf.level = 0.95)
```

```
##
## One Sample t-test
##
## data: y
## t = -0.59574, df = 24, p-value = 0.7215
## alternative hypothesis: true mean is greater than 100
## 95 percent confidence interval:
##  93.95993      Inf
## sample estimates:
## mean of x
##    98.44
```

Question 2: Political Economy

```
expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2026/main/datasets/expendi
```

1.

quick overview

```
head(expenditure)
```

```
##   STATE Y   X1 X2 X3 Region
## 1    ME 61 1704 388 399      1
## 2    NH 68 1885 272 598      1
## 3    VT 72 1745 397 370      1
## 4    MA 72 2394 458 868      1
## 5    RI 62 1966 157 899      1
## 6    CT 91 2817 162 690      1
```

```
str(expenditure)
```

```
## 'data.frame':   50 obs. of  6 variables:
## $ STATE : chr  "ME" "NH" "VT" "MA" ...
## $ Y : int  61 68 72 72 62 91 120 99 70 82 ...
## $ X1 : int  1704 1885 1745 2394 1966 2817 2685 2521 2127 2184 ...
## $ X2 : int  388 272 397 458 157 162 494 153 152 187 ...
## $ X3 : int  399 598 370 868 899 690 728 826 656 674 ...
## $ Region: int  1 1 1 1 1 1 1 1 1 2 ...
```

```
summary(expenditure)
```

```
##      STATE              Y              X1              X2
## Length:50          Min.   : 42.00      Min.   :1053      Min.   :111.0
## Class :character    1st Qu.: 67.25      1st Qu.:1698      1st Qu.:187.2
## Mode  :character    Median : 79.00      Median :1897      Median :241.5
##                               Mean  : 79.54      Mean  :1912      Mean   :281.8
##                               3rd Qu.: 90.00      3rd Qu.:2096      3rd Qu.:391.8
##                               Max.   :129.00      Max.   :2817      Max.   :531.0
##      X3              Region
## Min.   :326.0      Min.   :1.00
## 1st Qu.:426.2      1st Qu.:2.00
## Median :568.0      Median :3.00
## Mean   :561.7      Mean   :2.66
## 3rd Qu.:661.2      3rd Qu.:3.75
## Max.   :899.0      Max.   :4.00
```

loading libraries

```
library(ggplot2)
```

```
vars <- c("Y", "X1", "X2", "X3")
```

```
labels <- c(
  "Expenditure",
  "Income",
  "Financially Insecure",
```

```

    "Urban"
  )

par(mfrow = c(2, 3))

for (i in 1:(length(vars) - 1)) {
  for (j in (i + 1):length(vars)) {

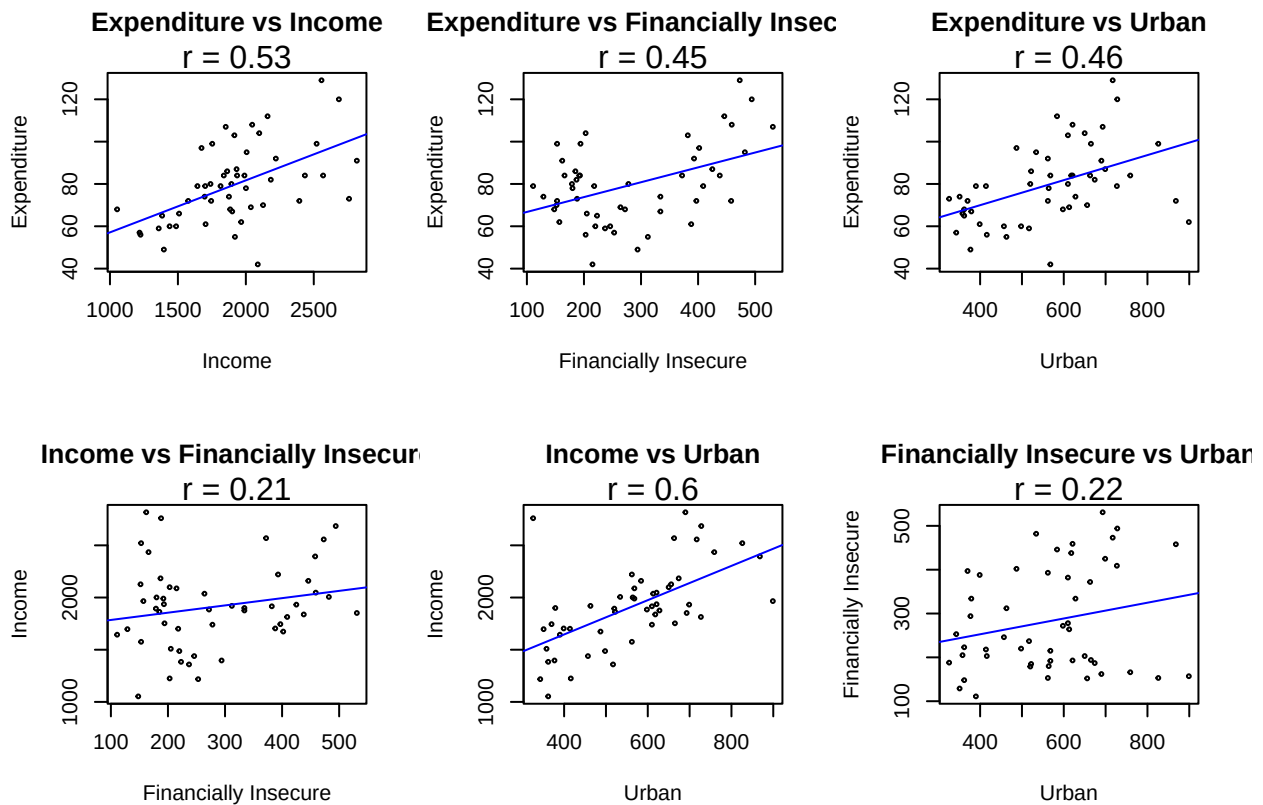
    plot(
      expenditure[[vars[j]]],
      expenditure[[vars[i]]],
      xlab = labels[j],
      ylab = labels[i],
      main = paste(labels[i], "vs", labels[j]),
      cex = 0.4
    )

    abline(
      lm(expenditure[[vars[i]]] ~ expenditure[[vars[j]]]),
      col = "blue",
    )

    r <- cor(
      expenditure[[vars[i]]],
      expenditure[[vars[j]]]
    )

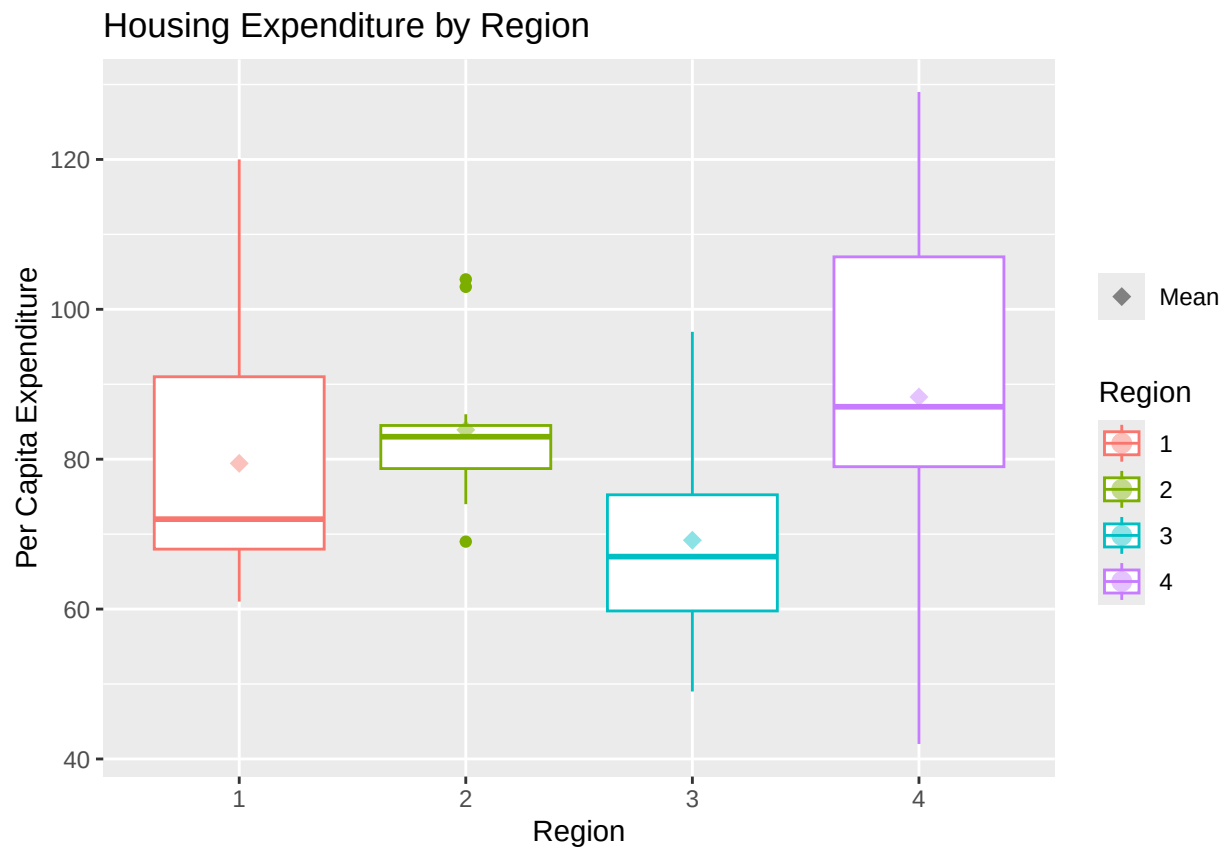
    mtext(
      paste("r =", round(r, 2)),
    )
  }
}

```



2.

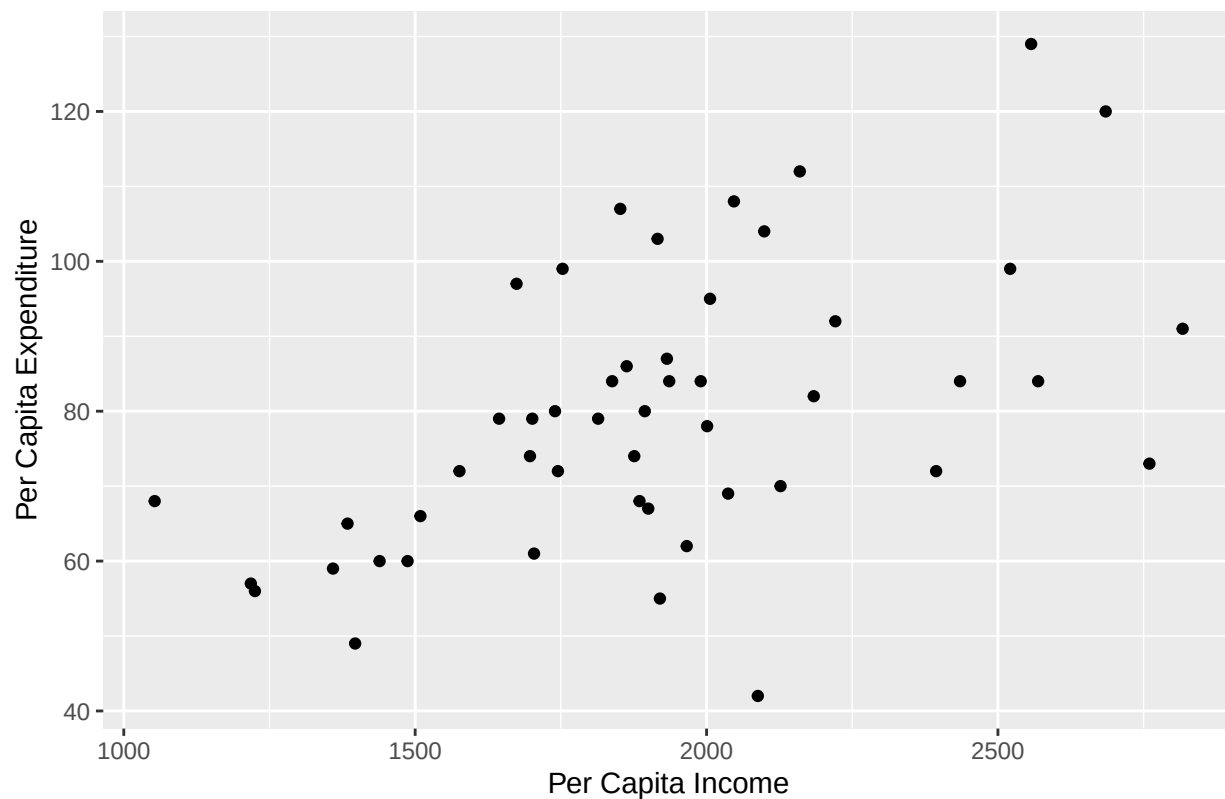
```
ggplot(expenditure, aes(x = factor(Region), y = Y, color = factor(Region))) +
  geom_boxplot() +
  stat_summary(
    aes(shape = "Mean"),
    fun = mean,
    geom = "point",
    size = 3,
    alpha = 0.45
  ) +
  labs(
    title = "Housing Expenditure by Region",
    x = "Region",
    y = "Per Capita Expenditure",
    color = "Region",
    shape = ""
  ) +
  scale_shape_manual(
    values = c("Mean" = 18)
  )
)
```



3.

```
ggplot(expenditure, aes(x = X1, y = Y)) +  
  geom_point() +  
  labs(  
    title = "Housing Expenditure vs Income",  
    x = "Per Capita Income",  
    y = "Per Capita Expenditure"  
  )
```

Housing Expenditure vs Income



```
ggplot(  
  expenditure,  
  aes(  
    x = X1,  
    y = Y,  
    color = factor(Region),  
    shape = factor(Region)  
  )  
) +  
  geom_point() +  
  labs(  
    title = "Housing Expenditure vs Income by Region",  
    x = "Per Capita Income",  
    y = "Per Capita Expenditure",  
    color = "Region",  
    shape = "Region"  
  )  
)
```

